Toward Personalized Hebrew ASR: Adapting a General Model to a Single Speaker

Dolev Abudi
Tel Aviv University
dolevabudi@mail.tau.ac.il

Hadas Yonat
Tel Aviv University
hadasyonat@mail.tau.ac.il

Abstract

Automatic speech recognition (ASR) systems are evaluated by error rates averaged across many speakers, which says little about how well they serve any one person, yet many deployments serve a single speaker. We ask two questions: does a Hebrew ASR model serve all its speakers equally, and how can it be adapted to a single speaker? On 261 speakers in Knesset committee recordings, a Hebrew fine-tune of Whisper lowers every speaker's error. The gain varies between speakers, and somewhat between groups such as fast and slow speakers, but speakers largely keep their order. We then adapt the Hebrew model to 12 speakers with LoRA. With 80 minutes of a speaker's speech, the adapter lowers the word error rate by a median 12%, and an adapter trained on the same amount of other speakers' speech lowers it by 10%, so the speaker's own speech adds 2–3%. We also train four speakers on up to 6 hours of their speech, and two of them on up to 24. Their adapters keep improving, up to a 34% lower word error rate than the Hebrew model, and a personal margin of up to 4% remains. Most of what adaptation buys is shared across speakers. Our code is available at https://github.com/hadasy-tau/deep-learning-project and our data at https://huggingface.co/knesset-asr.

1 Introduction

Modern ASR models such as Whisper (Radford et al., 2023) are evaluated by error rates averaged over many speakers, which can hide the speakers a model serves poorly. Yet many deployments serve one person, such as a Knesset member transcribed every day or the single user of a voice assistant, and what matters to them is how well the model recognizes their own voice. This calls for personalization: adapting a general model to one speaker with their own speech.

We study personalization for Hebrew and ask two questions: does a Hebrew ASR model serve all its speakers equally, and how can it be adapted to a single speaker? We answer them on Knesset committee recordings, which the Hebrew model was not trained on and which hold thousands of hours of speech from hundreds of identified speakers. Our contributions are threefold. First, a speaker-level error map of a general Whisper model and a Hebrew fine-tune over 267 speakers, the first for Hebrew. Second, adaptation of the Hebrew model to individual speakers with LoRA, on 5 minutes up to 24 hours of a speaker's speech, with the training settings tuned in 64 runs on validation data. The adapters lower the speakers' error, and to tell how much of this gain comes from the speaker's own speech, we compare each adapter with one trained on the same amount of other speakers' speech. Third, we find that most of what adaptation buys is shared across speakers: adapters keep improving with more data, and the speaker's own speech adds a margin over other speakers' speech that does not grow with more data. In practice, one adapter trained on speakers from the same setting can serve many speakers, without training a separate model for each. Training is cheap: an adapter on 80 minutes of speech trains in a few minutes on a standard GPU.

2 Related Work

Hebrew ASR. Whisper (Radford et al., 2023) is much weaker on languages with little training data, Hebrew among them. ivrit.ai narrowed this gap (Marmor et al., 2025) by fine-tuning it on Hebrew, mostly Knesset plenary speech (ivrit.ai, 2025), the starting point for our work.

Gaps within a language. Within a single language, too, ASR serves some people worse than others. Error rates differ by race (Koenecke et al., 2020), and by gender, age and accent (Feng et al., 2021). These studies compare groups, and we know of no such analysis for Hebrew. Adapting to a single typical speaker, Weninger et al. (2019) find gains that keep growing up to 20 hours of the speaker's speech, measured against the unadapted model.

3 Experimental Setup

3.1 Overview

Our study has two experiments. In the first, a general Whisper model and a Hebrew fine-tune (Section 3.3) transcribe the same speech, and we compare the error of each speaker and group under the two models. In the second, we select 12 speakers from this map: speakers the fine-tune helped least, speakers hard under both models, typical speakers, a speaker with a low word error rate (WER), and two speakers the fine-tune served well. We then adapt the Hebrew model to each speaker's own speech with LoRA, on 5 to 80 minutes and, for four speakers (two of the 12), up to 6 hours, and for two of them up to 24. We compare each adapter with a control trained on the same amount of speech from other speakers, which separates learning the voice from learning committee speech in general.

3.2 Data

We use `ivrit-ai/knesset-committees`, recordings of Israeli parliamentary committee sessions with their human-written protocols, force-aligned to the audio by a Whisper-family model so that every protocol word has a time span. Each protocol segment carries an alignment quality score, the median probability the aligner gives its words (Appendix A). The protocols are stenographic rather than verbatim, so a low score may reflect a mismatch between the text and the audio, and the protocol text is the reference for every WER we report (see Limitations). The corpus spans November 2018 to January 2026 with 13,177 sessions and 15,466 recorded hours, of which about 7,886 are aligned speech and about 7,070 have quality of at least 0.7.

3.3 Models

We compare two models. Arm **A** is `openai/whisper-large-v3` (Radford et al., 2023), the general multilingual model, and arm **B** is `ivrit-ai/whisper-large-v3` (ivrit.ai, 2025), the same model fine-tuned on roughly 4,700 hours of Hebrew *plenary* speech, so our committee recordings are new to it, though many of the speakers are not (see Limitations). Every adapter in this work is trained and evaluated on the full arm B. For the speaker-level analysis, arm B is a separate Hebrew fine-tune of OpenAI's turbo model with 4 decoder layers, ivrit.ai's `whisper-large-v3-turbo-ct2`.

3.4 Preprocessing

The released corpus needs two steps before use: a speaker-level analysis needs every protocol segment linked to a verified person, and Whisper takes at most 30 seconds of audio while sessions last hours. Like VoxKnesset (Marmor et al., 2026), a 2,300-hour corpus of Knesset plenary speech with verified speakers, we identify speakers by their official Knesset ids. We do not use VoxKnesset's audio, which comes from the plenary sessions arm B was fine-tuned on.

For the first, we link each protocol speaker name to a Knesset member, through the speaker attribution of the Knesset Corpus (Goldin et al., 2025) where it exists and by exact match to a member seated that day otherwise, and discard names that do not match. The matched identity also gives each member's birth year, gender, origin, nationality and religion, which we use to group speakers. For the second, each identified speaker turn is cut into pieces of at most 30 seconds, which we call chunks, that never cross a speaker change, with the aligned protocol text as the reference. The result is 1,204,617 chunks (3,841 hours) from 330 current and former Knesset members in 10,905 sessions.

Chunks with alignment quality below 0.7 are excluded from all analyses, and training and validation data are further restricted to chunks of at least 0.95, following ivrit.ai's recommendation, that also pass a word-level check (Appendix A).

3.5 Speaker-Level Performance

Both models transcribe the same chunks: up to one hour of speech per speaker, spread across many of their sessions, 65,990 chunks (230 hours) from 267 speakers in all. Arm A runs through Hugging Face Inference Providers and arm B through a RunPod endpoint, and both are told the language is Hebrew, so that neither is scored on language detection. For each speaker and group, we report WER under both models, pooled over the speaker's chunks (Appendix B), the gain $\mathrm{WER}_A - \mathrm{WER}_B$, and the relative gain, $(\mathrm{WER}_A - \mathrm{WER}_B)/\mathrm{WER}_A$.

3.6 Personalization

Speakers and data. The sessions of each of the 12 selected speakers are split into training, validation and test sets that share no session, with the newest sessions held out for testing. Each speaker is trained on budgets of 5, 20 and 80 minutes of their own speech, nested and filled from their most recent training sessions first. Every model is scored on two test sets from the same held-out sessions (Appendix A): chunks at quality ≥ 0.95 (*clean test*, about 45 minutes per speaker) and all chunks at quality ≥ 0.7 (*full test*, a median of 90 minutes, depending on how much of the speaker's speech aligns well). The clean test keeps chunks whose protocol matches

the audio, so its reference is closest to what was said, and the full test checks that our results do not depend on a filter set by a Whisper-family aligner (Appendix A). For each speaker, every model is scored on the same chunks of a test set.

Adapter and training. For each speaker and budget, we train a LoRA adapter (Hu et al., 2022) on arm B: added low-rank weights of rank 8 on the query and value projections of every attention layer (3.9M trainable parameters), while the model's own weights stay frozen. Training runs for at most 400 steps, more beyond 80 minutes, with early stopping on the speaker's validation loss (Table 1). At 80 minutes, validation loss bottoms out after about two passes over the data, and every run stops well before the step cap (Figure 13, Appendix D).

Choosing the training settings. We tune on validation data only, in 64 runs, with four speakers, one of each kind (helped least, hard under both models, typical and low WER), at 5 and 80 minutes, and 20 minutes uses the 80-minute choice. A run is scored by how much it lowers validation loss relative to arm B, $(L_B - L_{\text{best}})/L_B$, with L_B arm B's validation loss and L_{best} the adapter's lowest. Tuning proceeds in three stages, each starting from the previous choice: the learning rate (10^{-4}, 3×10^{-4}, 10^{-3}), then the LoRA rank (8, 16) with Whisper's dropout (0, 0.1), then augmentation (SpecAugment, alone or with $\pm 10\%$ tempo changes, but no pitch or speed perturbation, which would change the voice). A change replaces the default (learning rate 3×10^{-4}, rank 8, no dropout, no augmentation) only if it raises the score by at least 0.01 on average and for three of the four speakers. Raising Whisper's dropout removes most of the adapter's improvement, while neither rank 16 nor augmentation improves the score by the required margin (Table 1, Appendix C).

Control. To separate what an adapter learns about the speaker from what it learns about the committee setting, such as rooms, microphones and vocabulary, the 12 speakers are split at random into two folds of six. For each fold and budget, a *control* adapter is trained with the same settings on the same number of minutes from the six speakers of the *other* fold, so it never heard the speakers it is scored on. We report the relative gain over arm B, $(\text{WER}_B - \text{WER}_{\text{adapted}})/\text{WER}_B$, and the **personal margin**, $(\text{WER}_{\text{control}} - \text{WER}_{\text{own}})/\text{WER}_B$, which equals the own gain minus the control gain. Every comparison between two models, including the personal margin, gets a 95% interval and a p-value from a paired bootstrap over test chunks (Appendix E), and we call a difference significant when $p < 0.05$. All budgets use seed 0, and at 80, 360 and 1,440 minutes seeds 1 and 2 repeat the own adapters and the controls, with the folds redrawn, to separate speaker-level effects from seed variance.

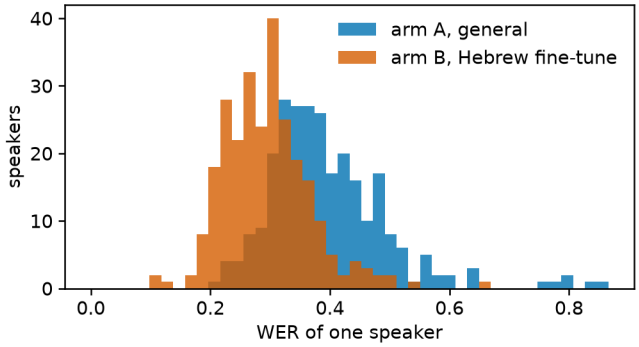


Figure 1: Per-speaker WER under arm A (general) and arm B (Hebrew fine-tune), all 267 speakers. Arm B shifts the whole distribution down, and its spread shrinks only in proportion to the lower average.

4 Results

4.1 Speaker-Level Error Map

After the quality filter, the analysis covers 267 speakers and 217 hours (58,180 chunks), with an overall WER, pooled over all chunks, of 0.387 for arm A and 0.292 for arm B. Statistics below are over the 261 speakers with at least 20 chunks, and per-speaker 95% confidence intervals come from bootstrapping each speaker's chunks (Appendix E).

The fine-tune helps every speaker, and speakers keep their order. All 261 speakers have a lower WER under arm B, by a median 24% of arm A's error (11% to 58%, Figure 6), and every improvement is beyond sampling noise (Appendix E). Yet the spread of per-speaker WER shrinks (standard deviation 0.099 to 0.069) only in proportion to the lower average (coefficient of variation 0.25 and 0.23, Figure 1). Speakers keep their order (Spearman $\rho = 0.90$ between WER_A and WER_B), and this ranking is stable across a speaker's recordings (Appendix D).

Group membership explains little. We compare the gain across seven groupings of speakers: gender, nationality, religion, religious orientation, country of origin, age quartile and speaking-rate tertile. The clearest difference is speaking rate (Figure 2, with the other six in Figure 11): slow speakers gain more than fast ones (27% vs. 22% of arm A's error removed). Arab, Muslim, Druze and Bedouin speakers also gain more than Jewish speakers (29–30% vs. 24%). But group sizes are highly imbalanced (224 Jewish speakers against 5 Bedouin),

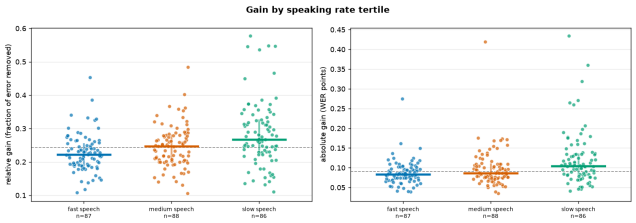


Figure 2: Gain from the Hebrew fine-tune by speaking-rate tertile, relative (left) and absolute (right). Each point is one speaker, the bar is the group median, and the dashed line is the overall median.

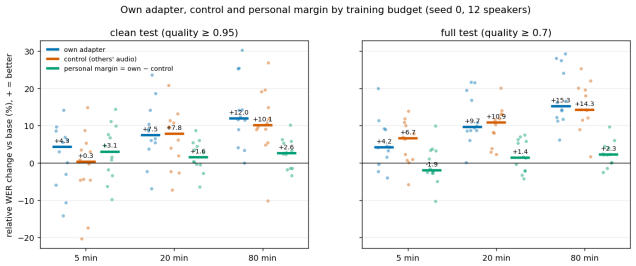


Figure 3: Relative WER change over arm B at 5, 20 and 80 minutes of training audio (seed 0, 12 speakers). Each dot is one speaker, and bars and numbers are medians. The median personal margin is the median of per-speaker differences.

and the variation within each group far exceeds the differences between group medians. A speaker's group thus says little about how much the fine-tune helps them, which motivates adapting to individual speakers.

4.2 Adaptation Results

The 80-minute adapter improves almost every speaker. With 80 minutes and seed 0, the own adapter lowers arm B's WER by a median 12.0% on the clean test and 15.3% on the full test (Figure 3, blue). It is better than arm B for 11 of 12 speakers on the clean test (9 significantly) and for all 12 on the full test (all significantly). The other two seeds agree: over all 36 speaker–seed pairs, it beats arm B in 35 on the clean test and in all 36 on the full test, and averaged over the three seeds, the clean-test gains range from +3% to +28% (median +11.4%, Figure 4). Figure 9 (Appendix D) shows the absolute WERs behind these gains.

Personal versus shared adaptation. The own adapter's median relative gain over arm B grows from +4.3% to +7.5% to +12.0% on the clean test at 5, 20 and 80 minutes, and from +4.2% to +9.7% to +15.3% on the full test (Figure 3). The control, which never heard the speaker, rises almost as fast, to +0.3%, +7.8% and +10.1% on the clean test and +6.7%, +10.9% and +14.3% on the full test. Most of the adaptation gain is therefore shared across speakers, and the speaker's own audio adds a median personal margin of +2.6% and +2.3% at 80 minutes, which stays between +1.6% and +3.1% on the clean test at every budget. No 5- or 20-minute adapter is the best model for any speaker (Figure 4), and at 5 minutes both adapters are unreliable: they are worse than arm B for some speakers, and the personal margin has opposite signs on the two test sets.

The margin differs between speakers, and it is not larger for the speakers the fine-tune helped least: it is largest for 23558 (David Bitan), at +9% to +10% on the full test in every seed, and absent for two of those three (Figure 12, Appendix D).

Beyond 80 minutes. To test whether the personal margin is limited by the amount of speaker data, we train four speakers on up to 360 minutes (6 hours), and two of them on up to 1,440 minutes (24 hours): 23558 (David Bitan), whose margin is the largest, and 23641 (Yaakov Asher), a speaker outside the 12 with one of the highest WERs. Each budget has its own control (Figures 5 and 10, Appendix D). The own adapters keep improving with diminishing returns, up to a 34% lower WER than arm B. The control improves as fast, so the added gain is shared across speakers, and the personal margin does not grow. It shrinks as a share of the gain but does not vanish, and at 24 hours it is 2–4%. Raising the LoRA rank from 8 to 32 does not change this conclusion (Appendix C).

5 Conclusion

Personalizing Hebrew ASR works, and it needs less than it seems. A general Hebrew fine-tune leaves the gaps between speakers in place, but a small LoRA adapter cuts a speaker's errors substantially, by up to 34%. Most of that gain does not require the speaker's own voice: speech from other speakers in the same setting delivers it, and the speaker's own recordings add a smaller margin on top. A practical system can therefore start from one adapter per setting and refine it per speaker, and keep collecting speech from that setting, since more data keeps improving a speaker's model even when the personal margin does not grow. Future work could adapt on similar rather than random speakers, and design training objectives that separate the voice from the setting.

Limitations

Prior exposure. Arm B was fine-tuned on Knesset plenary speech, which includes many of our speakers. Our evaluation uses committee recordings, a different setting and register, but the model may still be partly familiar with these voices.

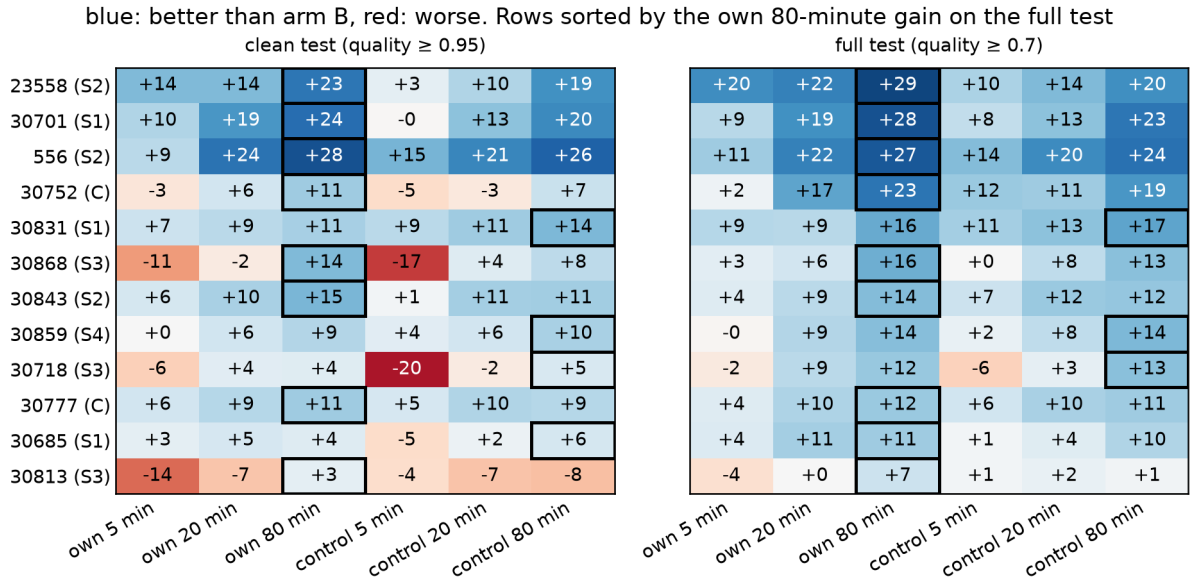


Figure 4: Relative WER change over arm B (%) for each speaker under each adapter, with the speaker's lowest WER boxed. The 80-minute columns average three seeds.

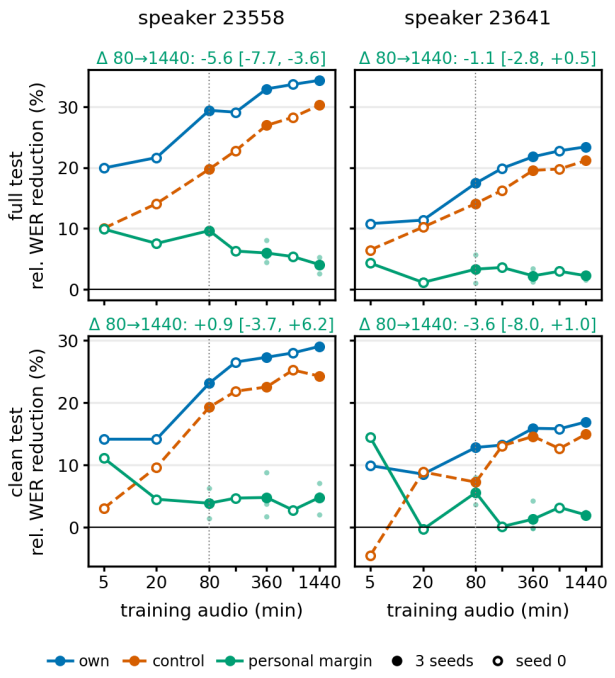


Figure 5: Relative WER reduction over arm B for the two speakers trained on up to 1,440 minutes. Filled markers average three seeds, hollow markers are seed 0, and the dotted line marks 80 minutes. Above each panel: the change in the personal margin from 80 to 1,440 minutes, with its 95% paired-bootstrap interval.

Shared recording conditions. Train and test sets are session-disjoint but not committee-disjoint: 75–82% of the test speech comes from committees the speaker also appears in during training. Part of the personal margin may therefore reflect familiar rooms and microphones rather than the voice.

The reference is the protocol. Every WER is measured against the stenographic protocol, which is not a verbatim transcript: stenographers condense, rephrase and omit speech, so part of every measured error is a mismatch between protocol and audio rather than a recognition mistake. Our gains are gains in agreement with the protocol, and may partly reflect learning what the stenographer omits. We limit this by training and validating only on chunks whose protocol closely matches the audio (quality ≥ 0.95, Appendix A) and by scoring every model on two test sets (Section 3.6). No verbatim reference was available, so our comparisons are most reliable between models scored against the same references.

Limited compute. All experiments ran on paid cloud services (RunPod GPUs and Hugging Face inference), which limited the models, configurations and speakers we could explore.

Division of Work

We developed the idea together from the start, over several brainstorming meetings, and sharpened it in discussions with the ivrit.ai team, who pointed us to problems worth studying. We set the guidelines of each stage together and worked on both stages jointly. Hadas carried out most of the in-depth speaker-level analysis, and Dolev focused on building the training infrastructure, with each of us closely involved in the other's part. We met regularly, in person and over Zoom, to make the project's decisions together, and we enjoyed the collaboration.

References

Siyuan Feng, Olya Kudina, Bence Mark Halpern, and Odette Scharenborg. 2021. Quantifying bias in automatic speech recognition. *Preprint*, arXiv:2103.15122.

Gili Goldin, Nick Howell, Noam Ordan, Ella Rabinovich, and Shuly Wintner. 2025. The Knesset corpus: An annotated corpus of Hebrew parliamentary proceedings. *Language Resources and Evaluation*, 59:2973–3004.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

ivrit.ai. 2025. ivrit-ai/whisper-large-v3. Hugging Face model card.

Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R. Rickford, Dan Jurafsky, and Sharad Goel. 2020. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689.

Yanir Marmor, Yair Lifshitz, Yoad Snapir, and Kinneret Misgav. 2025. Building an accurate open-source Hebrew ASR system through crowdsourcing. In *Proc. Interspeech 2025*, pages 723–727.

Yanir Marmor, Arad Zulti, David Krongauz, Adam Gabet, Yoad Snapir, Yair Lifshitz, and Eran Segal. 2026. VoxKnesset: A large-scale longitudinal Hebrew speech dataset for aging speaker modeling. *Preprint*, arXiv:2603.01270.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR.

Felix Weninger, Jesús Andrés-Ferrer, Xinwei Li, and Puming Zhan. 2019. Listen, attend, spell and adapt: Speaker adapted sequence-to-sequence ASR. In *Proc. Interspeech 2019*, pages 3805–3809.

Acknowledgments

We thank the ivrit.ai team (`https://www.ivrit.ai`) for their support throughout this project. Their ideas, practical advice and lessons from their own work on Hebrew speech helped shape our direction, and their open models and datasets made this study possible.

A Data Preparation Details

Alignment quality score. The corpus aligns each protocol to its recording with a Whisper-family forced aligner, which assigns every protocol word a time span and a probability: how likely the model finds that word at that point in the audio. A protocol segment's quality score is the median probability of its words. Our chunks of up to 30 seconds can join several protocol segments, so their score is the duration-weighted median over those parts. A low score means the text and the audio disagree, potentially because the stenographer condensed or omitted speech, or because of cross-talk. ivrit.ai treat 0.65 as fit for training and 0.95 as high quality.

Quality filter for the analysis. The 0.7 threshold removes 7,810 of 65,990 chunks (11.8%, 12.8 hours). Removing them lowers the overall WER from 0.417 to 0.387 (A) and from 0.324 to 0.292 (B), and leaves the difference between the arms almost unchanged.

Quality filter for training. Because the score is a median, a chunk can pass it while a few of its words are misaligned. Training and validation chunks therefore use the word probabilities directly and must pass four conditions: quality at least 0.95, at most 10% of words with probability below 0.5, no run of three or more such words, and the aligned words inside the chunk's time span must account for its reference text to within 20%, so that the checked probabilities cover the whole text.

Two test sets. The aligner is itself a Whisper-family model, so a 0.95 threshold may keep the chunks that Whisper models already transcribe well. Per chunk, the score predicts arm A's WER about as well as arm B's (Spearman $\rho = -0.62$ and -0.63 over the 65,990 chunks), so it does not favour the Hebrew fine-tune over the general model, and filtering on it does not tilt the comparison between them. It does mean that the filtered test set leans toward speech that every model we compare finds easy. We therefore score every model on two test sets drawn from the same held-out sessions: the filtered chunks, and all chunks with quality of at least 0.7.

B Implementation Details

Training. Adapters are trained with PyTorch, Hugging Face `transformers` (`Seq2SeqTrainer`) and `peft`, on the frozen base model in bfloat16. Each control's pool takes one chunk from each of its six speakers in turn until the budget is reached, and never from a session in which a speaker it is scored on is tested or validated.

Scoring. Reference and hypothesis are normalized (niqqud and punctuation removed) and aligned word by word, and substitutions, deletions and insertions are counted per chunk. WER is pooled over a speaker's test set: the sum of word errors divided by the sum of reference words. Confidence intervals and p-values come from a paired bootstrap over test chunks (Appendix E).

C Training Hyperparameters

Hyperparameter	Value
Base model	`ivrit-ai/whisper-large-v3`, frozen, bfloat16
Adapter	LoRA on the query and value projections of every attention layer (encoder, decoder self- and cross-attention)
Rank / α / LoRA dropout	8 / 16 / 0.05
Trainable parameters	3,932,160
Learning rate	10^{-4} at 5 minutes, 3×10^{-4} from 20 minutes on
Schedule	constant after 20 warm-up steps
Batch size	8, no gradient accumulation
Weight decay	0.01
Whisper dropout	0
Augmentation	none
Training length	at most 400 steps up to 80 minutes, 1,200 at 160 and 360, 3,000 at 720 and 1,440
Validation	loss every 20 steps
Early stopping	after 4 validations without improvement, best checkpoint restored, and not before one full pass at 720 and 1,440 minutes
Training labels	protocol text
Seeds	0 at every budget, 1 and 2 at 80, 360 and 1,440 minutes

Table 1: Training hyperparameters, shared by each own adapter and its control (the check in Table 2 uses rank 32, α 64).

Adapter capacity. To check whether a rank-8 adapter is too small to hold the voice, we repeat the 1,440-minute own adapters of 23558 and 23641 and their control at rank 32. The larger adapter does not make the personal margin grow: it rises in some cases and falls in others (Table 2).

Speaker	Test	Rank 8	Rank 32
23558	full	+2.6	+4.4
23558	clean	+7.1	+2.9
23641	full	+1.5	+2.2
23641	clean	+1.7	+6.7

Table 2: Personal margin (%) at 1,440 minutes, seed 0, with a rank-8 and a rank-32 adapter.

D Additional Results

D.1 Speaker-Level Error Map

Reliability of the per-speaker measures. Each speaker is measured on about one hour of speech, so their WER and gain carry sampling noise. To test whether the differences between speakers are a stable property of the speakers rather than of the particular

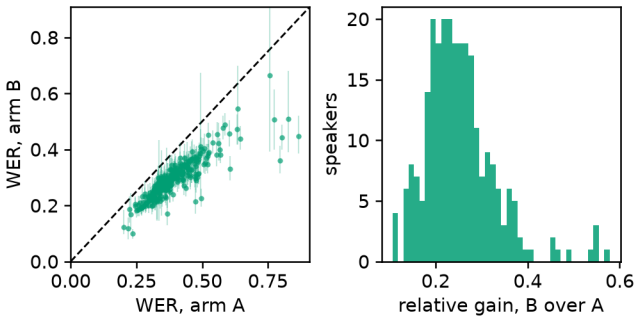


Figure 6: Left: per-speaker WER under arm A against arm B. Every point is one of all 267 speakers, the bar is the 95% bootstrap interval on WER_B alone, not on the gain, and the dashed line is no change. Right: distribution of the relative gain of arm B over arm A, for the 261 speakers with at least 20 chunks.

sample, we split each speaker's sessions into two alternating halves, compute WER_B and the relative gain on each half independently, and compare the halves across speakers (Figure 7). If the differences were noise, the two halves would be unrelated, and if they are stable, the points follow the diagonal. They do: the correlation between halves, corrected to full length (Spearman–Brown), gives a reliability of 0.78 for WER_B and 0.75 for relative gain. This is what justifies ranking speakers by their gain when selecting them for personalization.

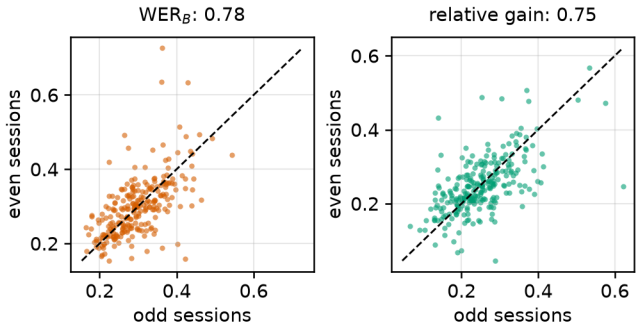


Figure 7: Each point is one speaker, measured separately on their odd and even sessions. Points near the diagonal mean the measure repeats across recordings, and titles give the split-half reliability (Spearman–Brown corrected).

Gain by group. Figure 11 shows the gain for the six groupings not shown in the main text (speaking rate is Figure 2). Speakers with missing metadata and groups of fewer than three speakers are omitted.

WER by year. As a check on the data, Figure 8 shows both models' WER by the year of the session. The two models follow the same trend across years.

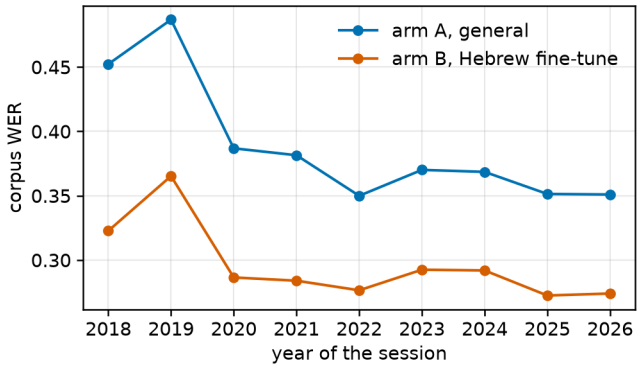


Figure 8: Overall WER of both models by the year of the session, after the quality filter.

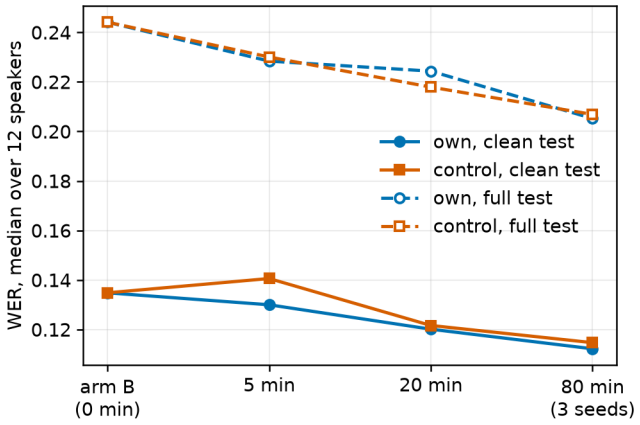


Figure 9: Median WER over the 12 speakers at 0 (arm B), 5, 20 and 80 minutes of training audio. At 80 minutes, each speaker's WER is averaged over three seeds.

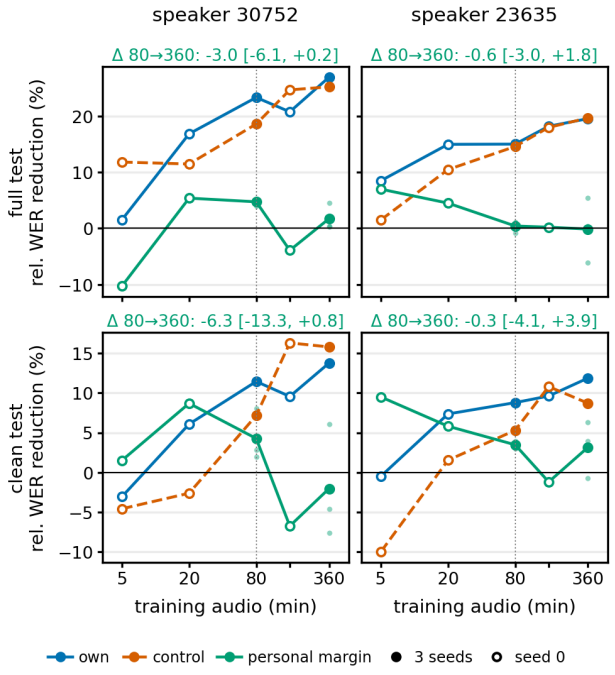


Figure 10: As Figure 5, for 30752 (one of the 12) and 23635 (from outside them), trained on up to 360 minutes.

D.2 Personalization

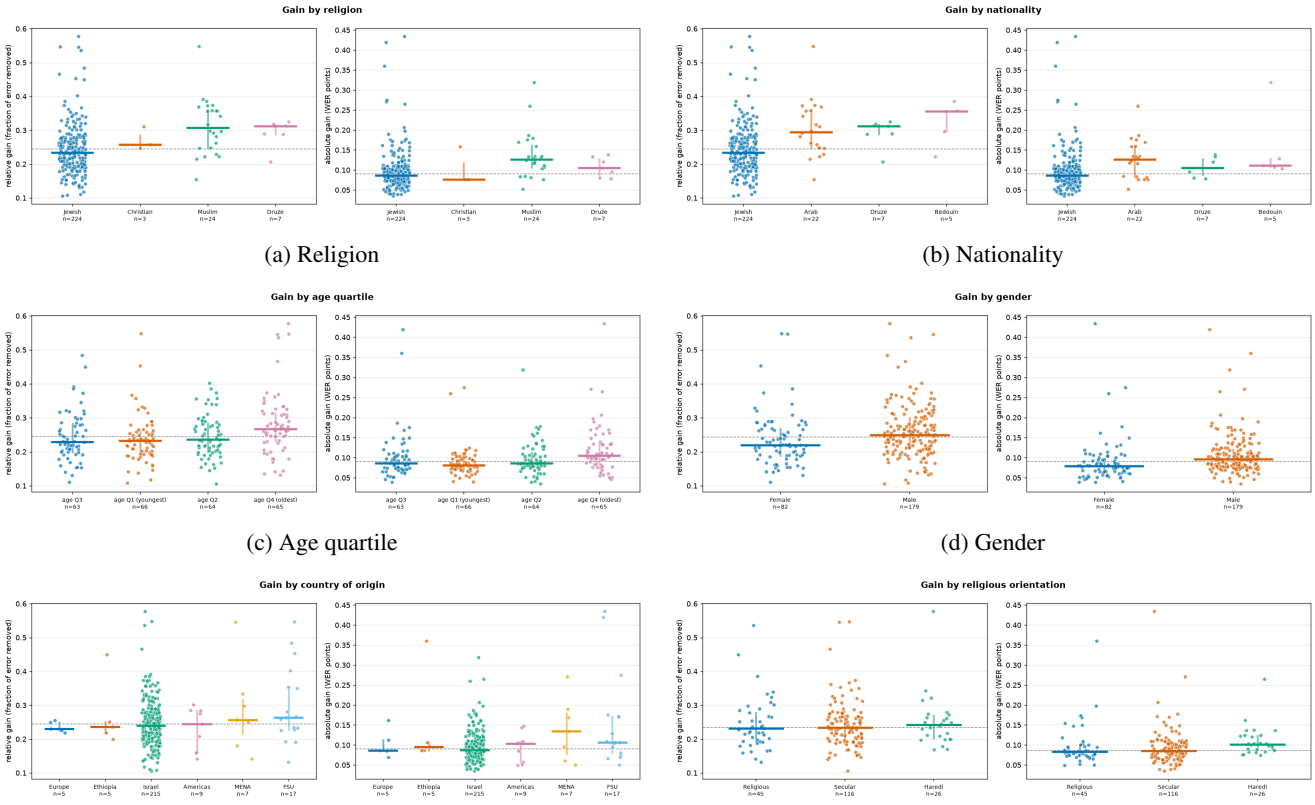
The figures below add detail to Sections 3.6 and 4.2: the WERs behind the gains (Figure 9), the data curve of the two speakers trained on up to 360 minutes (Figure 10), the personal margin per speaker and seed (Figure 12), and the learning curves of the 80-minute adapters (Figure 13).

E Bootstrap Intervals and Significance

Every interval and p-value in this paper comes from bootstrapping a speaker's chunks. A resampled set draws the speaker's chunks at random with replacement until it is the size of the original, every model in a comparison is scored on the same resampled set, and each set gives pooled WERs (the sum of word errors divided by the sum of reference words) and their difference. The 95% interval is the 2.5th to 97.5th percentile of the resampled differences.

Error map. We draw 1,000 sets per speaker, for WER_A, WER_B and the gain $\mathrm{WER}_A - \mathrm{WER}_B$. No p-value is computed: a speaker's improvement under arm B is beyond sampling noise when the interval of the gain lies entirely above zero, as it does for all 261 speakers with at least 20 chunks.

Adaptation. We draw 2,000 sets of each speaker's test chunks, for the gain of an adapter over arm B, $\mathrm{WER}_B - \mathrm{WER}_{\mathrm{adapted}}$, and for the personal margin, $\mathrm{WER}_{\mathrm{control}} - \mathrm{WER}_{\mathrm{own}}$. Dividing by WER_B gives the relative units of Section 4.2. The p-value is two-sided, since an own adapter could also do worse than its control: it is twice the share of the differences that fall on the far side of zero.



(a) Religion

(b) Nationality

(c) Age quartile

(d) Gender

(e) Country of origin

(f) Religious orientation

Figure 11: Gain from the Hebrew fine-tune by group, relative (left of each panel) and absolute (right). Each point is one speaker with at least 20 chunks. The bar is the group median, the line the interquartile range, and the dashed line the overall median.

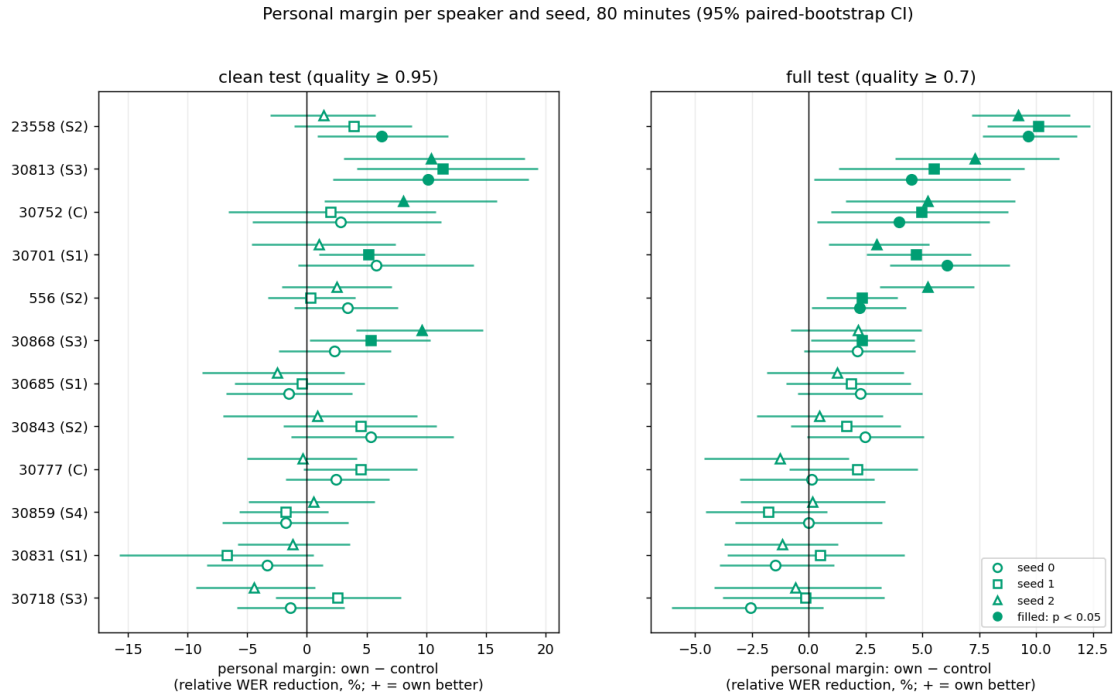


Figure 12: Personal margin per speaker and seed at 80 minutes, with 95% paired-bootstrap intervals. Filled markers are significant ($p < 0.05$). Profiles: S1, helped least by the fine-tune, S2, hard under both models, S3, typical, S4, low WER, and C, served well by the fine-tune.

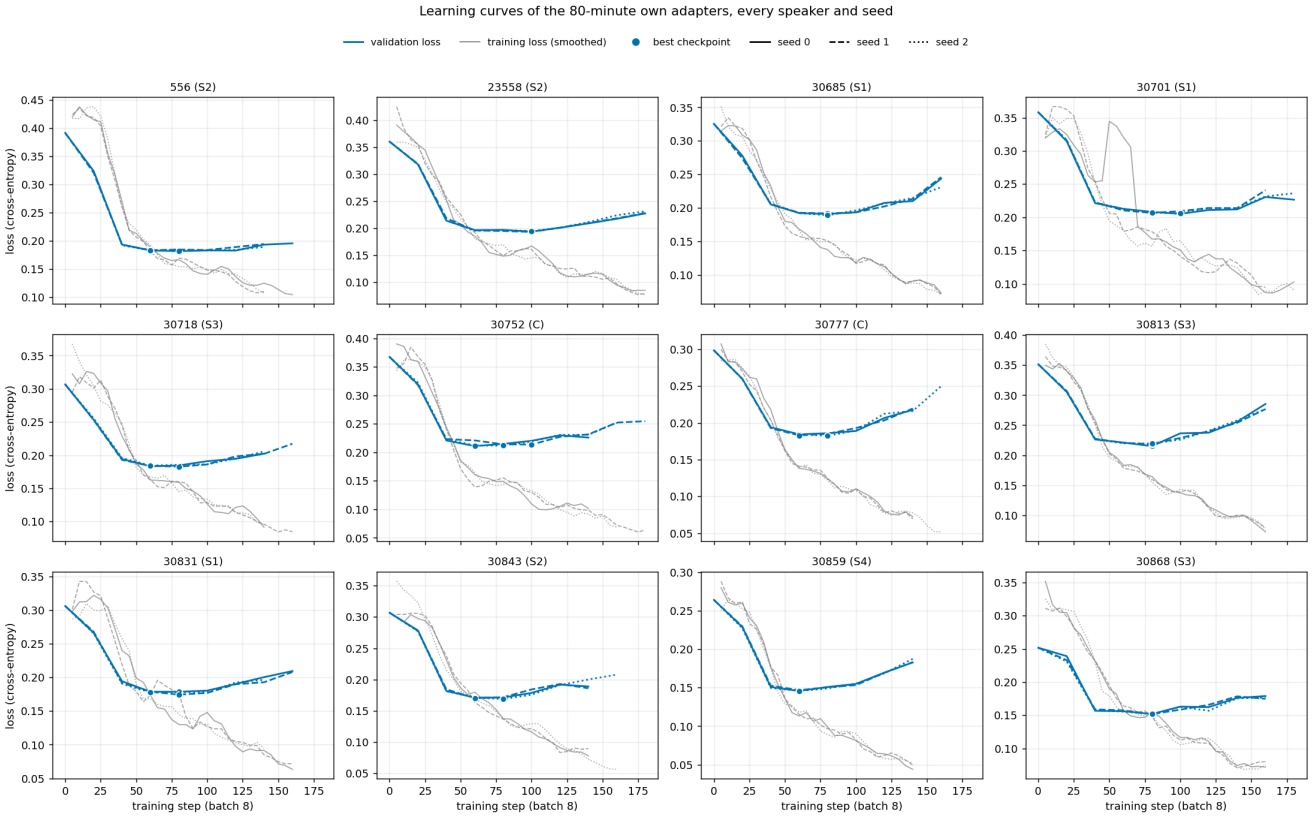


Figure 13: Training and validation loss of the 80-minute own adapters, one panel per speaker. Step 0 is arm B.